From a model to a measurement.

Every term you will hear today, defined for someone arriving cold. Yesterday you ran a model.
Today you defend a number. Keep this open beside the slides.

WORKSHOP MATERIALS
GITHUB.COM/JACQPARK/
INSTATS-LLM-WORKSHOP

THE ONE TO MEMORIZE · COHEN'S KAPPA

Cohen's kappa is the single best one-number summary of how well your model agrees with a human coder. Plain accuracy flatters you, because two coders labeling at random would still agree some of the time by luck. Kappa subtracts that luck. It takes the agreement you **observed**, removes the agreement you would **expect by chance**, and rescales so 1.0 is perfect and 0 is no better than guessing. Report it on your thirty-paragraph gold set and your methods reviewer will recognize it on sight.

$$\kappa = \frac{p_o - p_e}{1 - p_e}$$

p_o observed agreement, the share of paragraphs where model and coder match.
p_e agreement expected by chance, summed over classes from each side's label shares.

OBSERVED
Model and coder give the same score on 22 of 30 paragraphs. Raw accuracy is 0.73.

EXPECTED BY CHANCE
Given each side's class shares, luck alone would land about 0.41 of them.

KAPPA
(0.73 - 0.41) / (1 - 0.41) = 0.54. Moderate. 0.6+ is substantial, 0.8+ almost perfect.

THE BIG IDEA

Measurement
Assigning numbers to cases in a defensible way. A demo just classifies. A measurement can be defended to a reviewer.

Validity
The number tracks the concept it claims to. "Stance on climate" must not quietly become "general positivity."

Reliability
Re-running on the same input gives the same answer. Two coders on one paragraph reach the same score.

Comparability
Scores line up across countries and years with no drift introduced by the model itself.

Proxy
A measurable stand-in for a concept you cannot observe directly. Your new CSV column is a proxy.

THE CORPUS & THE RUN

Stratified sample
A sample drawn to keep balanced coverage, here across years, so a temporal trend is not distorted.

Context window *512 tokens*
The most tokens a model reads at once. DeBERTa's is 512, roughly 350 to 400 words.

Truncation
Cutting each text to the first 512 tokens before scoring. Document the choice in your appendix.

Batch *batch size*
How many texts the GPU scores at once. Sixteen here. Larger batches run faster on a T4.

ROLLING SCORES UP

Aggregation
Combining speech-level scores into a country-year number. The choice is never neutral.

Mean probability
Average the supportive-class probability across speeches. Keeps the continuous signal.

Share supportive
The fraction of speeches whose top label is supportive. Simpler, and it discards the probability.

Top-K
Keep only the most extreme K speeches per country-year. K is a hyper-parameter. Validate it against the gold set.

Sink label
The third "does not discuss climate" option. It absorbs off-topic speeches so they do not contaminate the proxy.

THE GOLD SET

Gold set *hand-coded*
The paragraphs you label by hand. The answer key you grade the model against. Thirty is the sweet spot.

Codebook
The written rules you fix before coding. Ours is three operational, disjoint, testable rules.

Inter-coder agreement
How often two human coders give the same paragraph the same score. Your human ceiling.

Class balance
How many cases fall in each label. Print it so a lopsided sample does not fool the accuracy.

Random seed
A fixed number that makes a random draw repeat exactly. We use seed 7 so the sample is reproducible.

GRADING THE MODEL

Confusion matrix
A table of model labels against gold labels. The diagonal is agreement, off-diagonal cells are the mistakes.

Accuracy
Share of paragraphs the model got right. It can mislead when the classes are imbalanced.

Weighted F1
Accuracy's more honest cousin. Balances precision and recall, then weights by class size.

Cohen's kappa
Agreement corrected for chance. Day 2's headline number. 0.4 to 0.6 is moderate.

Pearson r
Correlation between the supportive probability and the gold score. High r means sensible probabilities even when the label is wrong.

Spearman rho
The rank version of r. Robust to outliers and monotone shifts. Report both.

Calibration
Whether a 0.9 score really means about 90 percent likely right. Inspect the probability histogram.

COMPARABILITY CHECKS

Subgroup table
Kappa and accuracy computed separately by era and by region.

Drift
Performance that shifts across subgroups. A swing above 0.15 in kappa flags a comparability problem.

Face-validity check
Plotting the measure against a known event, the Paris Agreement here, to see if it behaves sensibly.

BEATING THE CEILING

Off-the-shelf ceiling
The best kappa a generic zero-shot model reaches with no training. Often 0.4 to 0.7 on this corpus.

Few-shot fine-tuning
Teaching a model your specific task from a handful of labeled examples.

SetFit
A method that fine-tunes a sentence embedder with contrastive pairs. Learns from as few as eight examples per class.

Cross-encoder
Reads your text and your label together and judges entailment. The DeBERTa-NLI setup. It never learns your task.

Bi-encoder
Turns each text on its own into one embedding vector. What a sentence-transformer is, and what SetFit fine-tunes.

Contrastive learning
Training that pulls same-label vectors closer together and pushes different-label ones apart.

Train / test split
Holding back part of your labels to score the fine-tune honestly. Two thirds to train, one third held out.

Hypothesis ensembling
Averaging several worded versions of one label. Needs no training data. Lifts kappa about 0.05 to 0.10.

Full fine-tuning
Hand-code 300 paragraphs and train a DeBERTa head. Lifts kappa 0.10 to 0.20 at a week's cost.

REPRODUCIBILITY & THE APPENDIX

Model ID
A model's permanent Hugging Face name with a version pin, e.g. `MoritzLaurer/DeBERTa-v3-base-mnli-fever-anli`.

Sidecar file
A small text file shipped beside the CSV that records the exact candidate-label list.

Replication package
The notebook, the gold-set CSV, and the data pointer a reviewer needs to rerun you.

Validation appendix
Where the codebook, kappa, confusion matrix, and subgroup table all live.

Limitations
Where truncation, label wording, and translation drift each get an honest paragraph.